

K-Means vs Agglomerative Hierarchical Clustering

Feature	K-Means Clustering	Agglomerative Hierarchical Clustering
Approach	Partitioning (Top-down)	Bottom-up merging
Working Principle	Divides into k clusters minimizing variance	Merges closest clusters step by step
Number of Clusters	Must specify k before	Decided after using dendrogram
Cluster Shape	Spherical/Convex	Arbitrary shapes
Output	Clusters + centroids	Dendrogram (tree)
Speed	Fast, suitable for large data	Slower for large datasets
Flexibility	Less flexible	More flexible
Outliers	Sensitive	Less sensitive
Decision Making	Elbow/Silhouette methods	Cut dendrogram
Determinism	Non-deterministic	Deterministic